

KineCAM: An Instant Camera for Animated Photographs

Ticha Sethapakdi
ticha@mit.edu
MIT CSAIL
Cambridge, USA

Mackenzie Leake
leake@mit.edu
MIT CSAIL
Cambridge, USA

Catalina Monsalve Rodriguez
catamon@mit.edu
MIT CSAIL
Cambridge, USA

Miranda J. Cai
mjcai@mit.edu
MIT CSAIL
Cambridge, USA

Stefanie Mueller
stefanie.mueller@mit.edu
MIT CSAIL
Cambridge, USA



Figure 1: *KineCAM* is an instant camera-inspired device that allows users to capture and fabricate animated photographs, in the form of kinegrams, on the spot.

ABSTRACT

The kinegram is a classic animation technique that involves sliding a striped overlay across an interlaced image to create the effect of frame-by-frame motion. While there are known tools for generating kinegrams from pre-existing videos and images, there exists no system for capturing and fabricating kinegrams in situ. To bridge this gap, we created *KineCAM*, an open source¹ instant camera that captures and prints animated photographs in the form of kinegrams. *KineCAM* combines the form factor of instant cameras with the expressiveness of animated photographs to explore and extend creative applications for instant photography.

ACM Reference Format:

Ticha Sethapakdi, Mackenzie Leake, Catalina Monsalve Rodriguez, Miranda J. Cai, and Stefanie Mueller. 2022. KineCAM: An Instant Camera for Animated Photographs. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3526114.3558621>

¹<https://github.com/HCIELab/KineCAM>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA
© 2022 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9321-8/22/10.
<https://doi.org/10.1145/3526114.3558621>

1 KINECAM DESIGN

KineCAM is a custom-built camera device (Fig. 2) with kinegram processing software written in Python. It retains many of the key qualities of instant photos and moving photos but differs in several ways. Similar to instant cameras, *KineCAM* offers a fast and portable way to create physical photos. Rather than capturing static images, however, *KineCAM* is able to capture dynamic movements, which expands the artistic potential for instant photography. Unlike the “black box” of a Polaroid camera, *KineCAM* also provides access to the internal hardware and software systems, which enables users to customize their cameras and produce a wider range of visual outputs.

1.1 Components and workflow

KineCAM’s workflow uses a three-tiered approach that encompasses capture, processing, and fabrication. When the photographer presses the shutter button, the indicator LED lights up for 1 second to show that *KineCAM* is capturing a video and turns off when it is finished. After extracting frames from the video and processing them into a kinegram, the Raspberry Pi prints the output on the thermal printer. The striped overlays are printed separately on transparency film using a regular inkjet printer.

1.2 Software implementation

To create the kinegram, our system records the video for a fixed unit of time and selects n frames from it. It then decomposes the frames into strips of width w pixels, interlaces them, and composites them into a single image. For an n -frame kinegram, its striped overlay

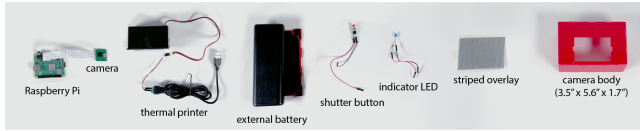


Figure 2: The core components of *KineCAM* are the Raspberry Pi 3+, Arducam 5MP Camera, and a small thermal receipt printer. These are off-the-shelf components that can be acquired through most electronics retailers and together cost less than \$100.

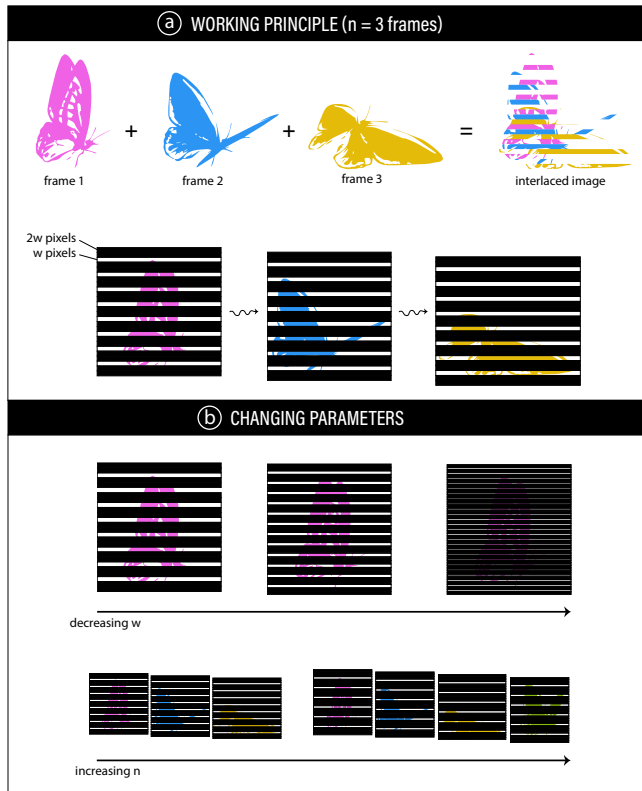


Figure 3: A system diagram showing a) the working principle for *KineCAM* and b) the effect of changing certain parameters.

comprises alternating transparent stripes of width w and opaque stripes of width $(n - 1)w$. Constructing the overlay in this way ensures that only one frame is visible at a time (Fig. 3a).

In our examples, we set the default video capture length to 1 second (32 frames total), create strips that are 3 pixels wide, and select frames near the beginning (frame 9), middle (frame 19), and end (frame 29). We picked these values because they produced subjectively good results for our purposes, but these parameters can be adjusted.

2 RESULTS

We used *KineCAM* to create a portrait series that captures an afternoon social gathering (Fig. 4). Subjects were invited to have their picture taken while we observed how they interacted with the camera. Our experiences from this activity enabled us to see how *KineCAM* creates opportunities for experimental, low-stakes, social photography.



Figure 4: Selected scans of the portrait series made with *KineCAM*.

3 CONCLUSION

While there is tremendous value in approaching technology with a forward-looking perspective, it can also be inspirational to look back and revisit technologies with a sense of nostalgia. By drawing inspiration from the instant cameras of decades past and the longstanding fascination with adding motion to photos, *KineCAM* creates new photography experiences that encourage play and experimentation. We hope that open-sourcing *KineCAM* can enable future artists, makers, and engineers to reappropriate *KineCAM* for other use cases and contexts that make it greater than the sum of its parts.

ACKNOWLEDGMENTS

We thank Jasmin Bouzarouata, Thanh Nguyen, and Jahrid Clyne for all their help.